

CONTIKI For Converting EMBEDDED Into The IoT

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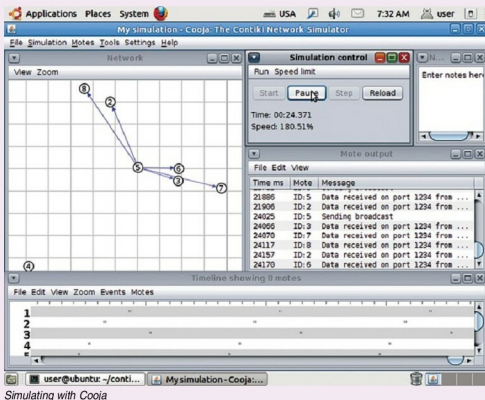
We are in an era of rapidly-developing electronics of embedded systems, efficient processors, the Internet of Things (IoT) and miniature chips. To aid with one's design process in this fast-paced scenario, here is an operating system (OS) that is specially designed to connect tiny, low-cost and low-power microcontrollers (MCUs) to the Internet—Contiki.

Contiki has been developed with an aim to connect embedded system designs to the Internet. The OS is customised to enable fast and easy development and inter-operate with most readily-available MCUs. The design is optimised for low-power applications, what with minimal power consumption being the driving factor behind most IoT applications today. This article tries to have a look at what Contiki offers for the IoT and for easy development.

Ease through the protocol maze

It is well-known that working with any IoT product demands adhering to stringent norms and protocols. While the idea of implementing these from scratch demands a lot of work and re-work from the developing team, it is a relatively smooth sail once the base is set. Contiki is modelled over Internet Protocol (IP) suite and offers support to the standard User Datagram Protocol (UDP), Transmission Control Protocol (TCP) and Hypertext Transfer Protocol (HTTP).

For simple applications like alerting your friend or for applications where bandwidth is premium, it might be wise to shy away from



Simulating with Cooja

Contiki in a nutshell

- An operating system for the IoT
- Developed by the team headed by Adam Dunkels of Thingsquare
- Can be freely used for both commercial and non-commercial purposes (BSD licence)
- Latest version - 3.0
- To get started, log on to www.contiki-os.org/start.html

elaborate or complex designs. For such situations, Contiki comes up with Rime, a tailored wireless networking stack that can extend to operations like address-free multi-hop. Rime stack works by offering a default set of communication primitives for low-power wireless systems, which can be executed on their own or combined with complex ones.

A low-power focus. With IoT designs focussing on reducing power consumption to bare-minimal

levels, this OS lets one work easily with new-age, low-power protocol standards. The recently-standardised Internet Engineering Task Force (IETF) protocols for low-power IPv6 networking, including IPv6 over Low-power Wireless Personal Area Network (6LoWPAN) adaptation layer, Routing Protocol for Low-power (RPL) IPv6 multi-hop routing protocol and Constrained Application Protocol (CoAP) RESTful application-layer protocol, are also supported.